MATLAB Tutorial

Basic Operations

- Math Operators +, -, *, /, ^ (power)
- Array, matrix

$x = [1, \quad 2, \quad 3]$

>> x=[1,2,3];

$x = [1, \quad 2, \quad 3]^T$

>> x=[1;2;3];

$$x = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

>> x=[1,2;3,4];

$$x = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$y_1 = [1 \quad 2]^T$

$y_2 = [3 \quad 4]^T$

```
>>  x=[1,2;3,4];
    y1=[1;2];y2=[3;4];
>>  y1+y2
ans =
    4
    6
```

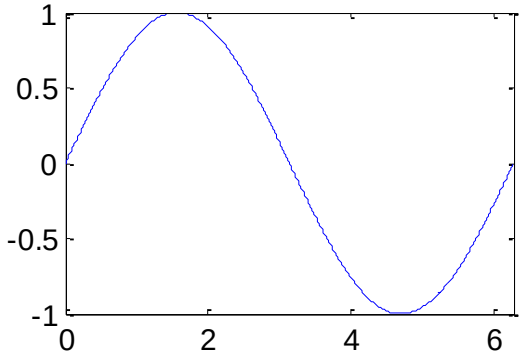
$y_1 + y_2 = [4 \quad 6]^T$

```
>> x*y1
ans =
    5
    11
```

$$xy = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

- Graph

```
>> t = 0:0.01:2*pi;
   y = sin(t);
   plot (t,y)
```



Creating A Transfer Function

- Command '**tf**' in MATLAB creates a transfer function.
- Given a numerator and a denominator express them in vector format:

  Example (second order system):

  Define: num= [2, 10]

  den = [1,2,10]

$$\frac{Y(s)}{U(s)} = \frac{2s+10}{s^2+2s+10}$$

- Define Sys = tf(num,den)

Other Useful Functions

Command	Description
C = conv(A, B)	Convolution and polynomial multiplication
p=roots(den)	Roots of the characteristic polynomial den=0
[r,p,k]=residue(num,den)	Calculates the partial fraction expansion of (num/den)
[z,p]=tf2zp(num,den)	z=roots of num=zeros of (num/den) p=roots of den =poles of (num/den)
[A,B,C,D] = tf2ss(num,den)	Transfer function to state-space conversion x = Ax + Bu y = Cx + Du
ilaplace	Inverse laplace transform
series	Series interconnection of the two LTI models
feedback	Feedback connection of two LTI models

Example

- Roots of $s^2 + 2s + 1 = 0$

```
>>roots([1 2 1])
   ans =   -1   -1
```

- [r,p,k]=residue(num,den)
- Example:

$$\frac{B(s)}{A(s)} = \frac{s^4 + 8s^3 + 16s^2 + 9s + 6}{s^3 + 6s^2 + 11s + 6}$$

$$\frac{B(s)}{A(s)} = k(s) + \frac{r(1)}{s - p(1)} + \frac{r(2)}{s - p(2)} + \ldots + \frac{r(n)}{s - p(n)}$$

Using Residue Command:

$$\frac{B(s)}{A(s)} = s + 2 + \frac{-6}{s+3} + \frac{-4}{s+2} + \frac{3}{s+1}$$

In MATLAB:

```
num=[1 8 16 9 6];
den=[1 6 11 6];
[r,p,k]=residue(num,den)
```

Example

- Expand $\left(s^2+2s+2\right)\left(s+4\right)\left(s+1\right)$

```
>>c=conv([1 2 2],conv([1 4],[1 1]))
```

c = 1 7 16 18 8 $\rightarrow s^4+7s^3+16s^2+18s+8$

- Inverse Laplace

$$Y(s)=\frac{2s+10}{s^2+2s+10}\cdot\frac{1}{s}$$

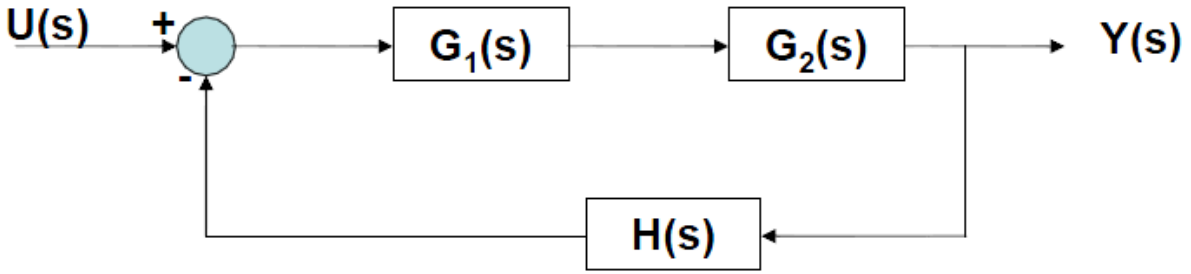
In MATLAB:

```
syms s
G=(2*s+10)/(s*(s^2+2*s+10))
g=ilaplace(G)
```

$$y(t)=1-e^{-t}\cos 3t+\frac{1}{3}e^{-t}\sin 3t$$

Block Diagram Manipulation

Example:



$$G_1(s) = 2s + 10$$

$$G_2(s) = \frac{1}{3s^2 + 9}$$

$$H(s) = \frac{1}{s + 0.2}$$

In MATLAB:

```
G1 = tf([2,10],[1])
G2 = tf([1],[3,0,9])
H = tf([1],[1,0.2])
G12=Series(G1,G2)
G=Feedback(G12,H)
```

Forced Response Commands

Command	Description
Step(sys) or **Step**(num,den)	Plots unit response of system ‘sys’
impulse(sys) or **impulse**(num,den)	Plots the unit impulse response of system ‘sys’
lsim(sys,U,T) or **lsim**(num,den,U,T)	Plots system response for arbitrary input U at time instant in vector T

Example

System

$$\frac{Y(s)}{U(s)} = \frac{2s+10}{s^2+2s+10}$$

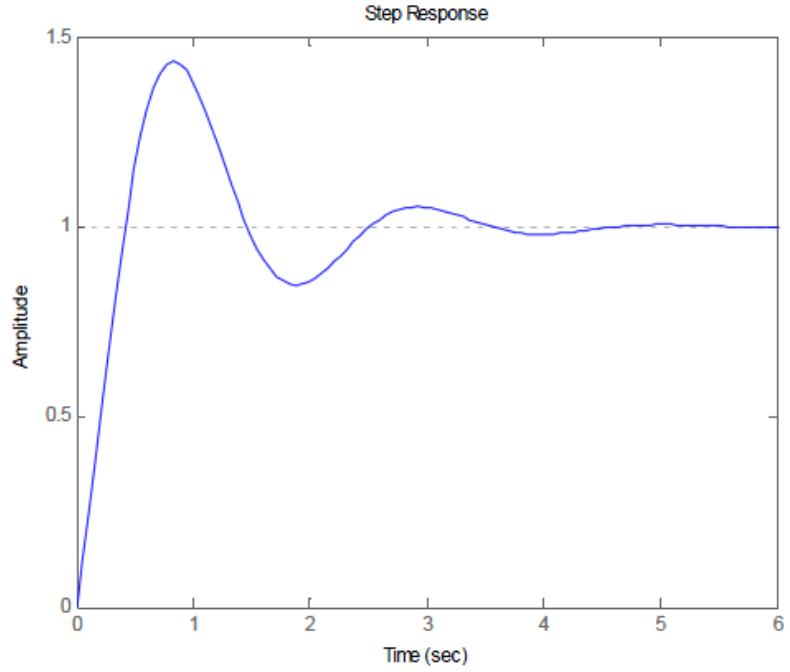
Step Response:

$$U(s) = \frac{1}{s}$$

$$Y(s) = \frac{2s+10}{s^2+2s+10}\frac{1}{s}$$

Example

- Step Response



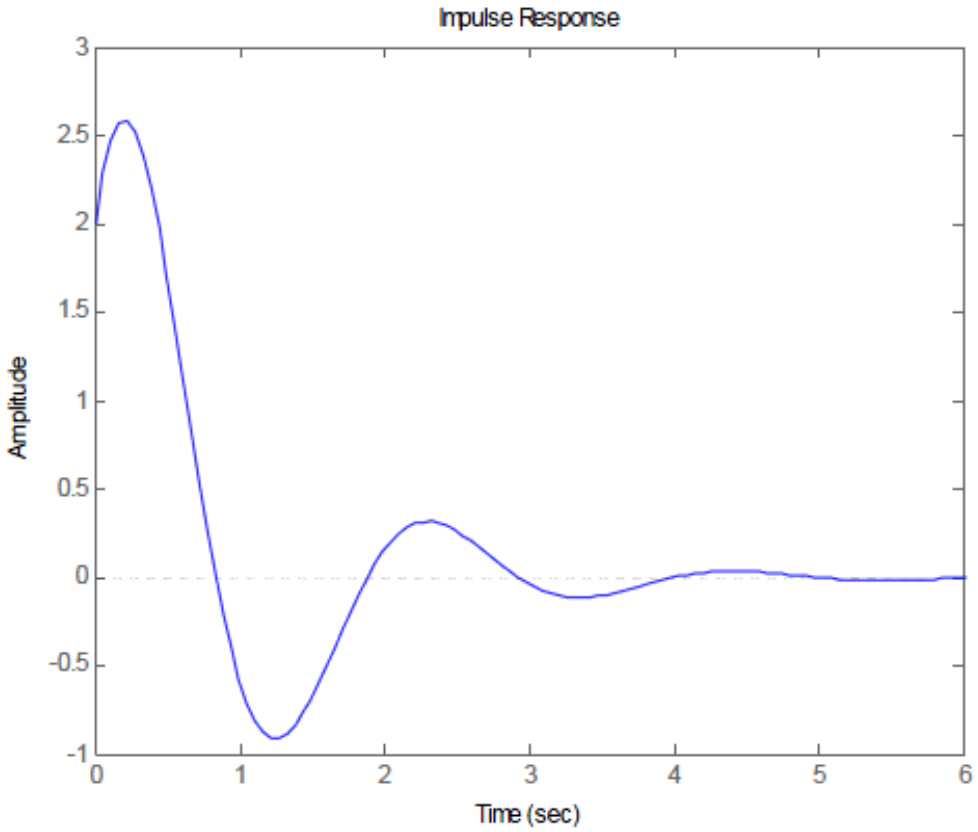
In MATLAB:

```
num=[2 10];
den=[1 2 10];
sys=tf(num,den)

%Step Response
figure(1)
step(sys);
```

Example

Impulse Response



In MATLAB:

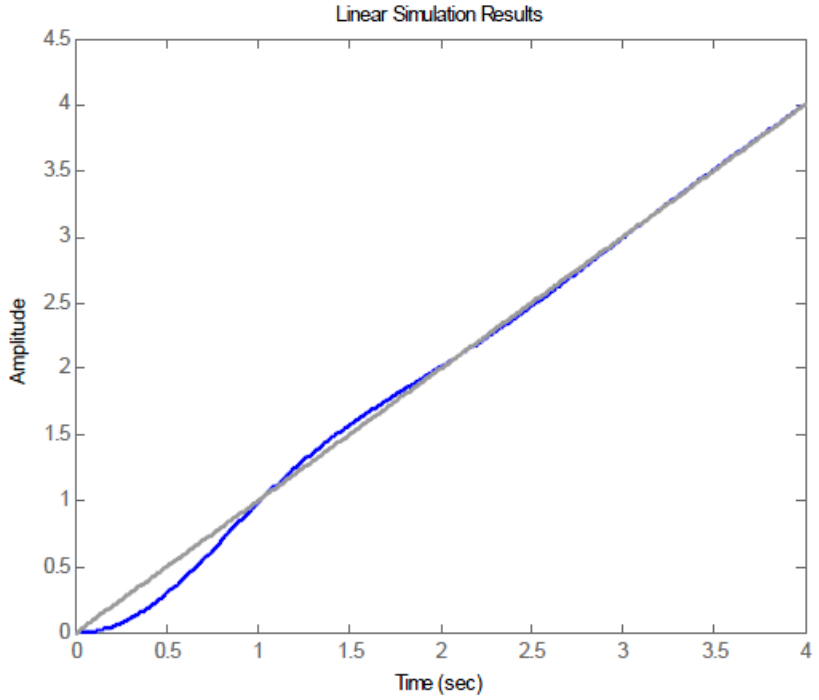
```
num=[2 10];
den=[1 2 10];
sys=tf(num,den)

%Step Response
figure(1)
impulse(sys);
```

Example

-- Arbitrary input

- Ramp Input (using '**lsim**')



In MATLAB:

```
num=[2 10];
den=[1 2 10];
sys=tf(num,den)

t=[0:0.01:4];
u=t;
figure(3)
lsim(sys,u,t)
```

Example

- Another way to do Ramp Input

$$\frac{Y(s)}{U(s)} = \frac{2s+10}{s^2+2s+10}$$

Ramp Input
$U(s) = 1/s^2$

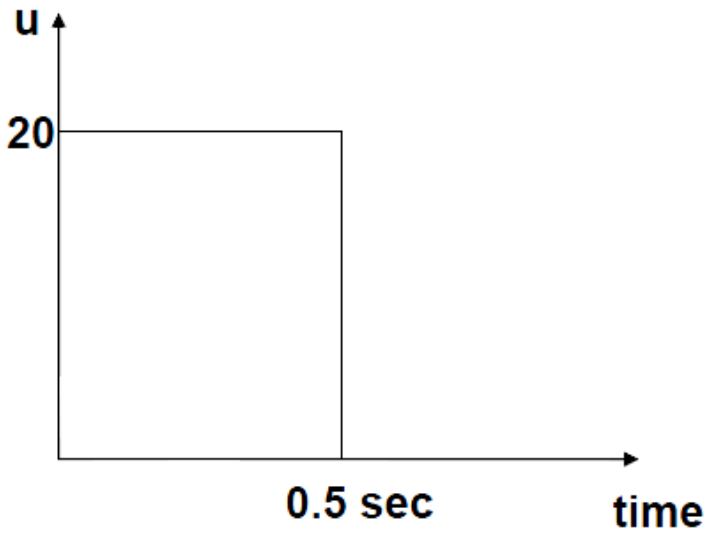
$$Y(s) = \frac{2s+10}{s^2+2s+10}\frac{1}{s^2}$$

$$Y(s) = \frac{2s+10}{s^3+2s^2+10s}\frac{1}{s}$$

Use Step function

Example

- Pulse Input (using '**lsim**')



In MATLAB:

```
num=[2 10];
den=[1 2 10];
sys=tf(num,den)

t=[0:0.01:5];
u=zeros(1,length(t));
u([1:50])=20;
figure(3)
lsim(sys,u,t)
```